THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 21	Each term is the sum of the two terms before it.
2	**(b)** 90	At full hours the minute hand is at 12; each hour mark is 30°.
3	**(d)** 55	Reverse and add.
4	**(d)** 4.5	Divide by the value of one symbol.
5	**(b)** 3516	Apply the divisibility rule of 4.
6	**(b)** 3	Number of lines $= \frac{3\times2}{2}$.
7	**(d)** acute	Classify the turn.
8	**(a)** 5175	Arrange digits in descending and ascending order and subtract.
9	**(a)** 8	Divide the number by the scale.
10	**(a)** 8 and 15	Co-prime numbers have no common factor other than 1.
11	**(c)** 25	Two consecutive triangular numbers add up to a square number.
12	**(c)** 75	Add the adjacent angles.
13	**(c)** 9	Even: halve it. Odd: multiply by 3 and add 1.
14	**(d)** 20	Subtract the smaller number from the larger.
15	**(d)** 11 and 13	Twin primes are primes that differ by 2.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Each term is the sum of the previous two (1); $1, 1, 2, 3, 5, 8, 13, 21, 34, 55$ (1).	1+1
17	Adjacent angles add up (1); $46 + 38 = 84^\circ$ (1).	1+1
18	47 each gets reversed and added (1); Palindrome reached: 121 after 1 step(s) (1).	1+1
19	Number of symbols = number $\div 5$ (1); Guava: 5 symbols; Mango: 2 symbols; Banana: 7 symbols (1).	1+1
20	Last two digits 16 give divisible by 4 (1); Last three digits 216: divisible by 8 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Sum $= 45 + 55 = 100$ (1); $100 = 10 \times 10$ (1); Two consecutive triangular numbers always add to a square number (1).	1+1+1
22	Every 90° is a quarter turn: N to E to S to W (1); 90° gives East (1); Type: right (1).	1+1+1
23	Repeat the largest-minus-smallest step (1); $3524 \to 3087 \to 8352 \to 6174$ (1); Number of steps $= 3$ (1).	1+1+1
24	(i) Banana: 40 (1); (ii) Apple: 10 (1); (iii) total $= 80$ (1).	1+1+1
	OR: Sum of digits at odd places and even places (1); Difference $= 11$ (1); Divisible by 11 (1).	1+1+1
25	$1 + 2 + \cdots + 3 = 6$ (1); $1^3 + \cdots + 3^3 = 1 + 8 + 27 = 36$ (1); $6^2 = 36$, so they match (1).	1+1+1
	OR: Draw a ray OA, place the centre of the protractor at O with the zero line along OA (1); Mark a point at 40° and join O to it (1); The angle is acute (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	First person shakes 5 hands, next 4 new hands, ... down to 1 (1); Total $= 5 + 4 + 3 + 2 + 1$ (1); $= 15$ handshakes (1); This is the triangular number T_5 (2).	1+1+1+2
	OR: $\angle AOC = 110 - 30 = 80^\circ$ (1); $\angle AOB = 30^\circ$: acute (1); $\angle BOC = 110^\circ$: obtuse (1); $\angle AOC = 80^\circ$: acute (1); State the arms and vertex (1).	1+1+1+1+1
27	Step 1: 4311-1134=3177 (1); Step 2: result 6354 (1); Step 3: result 3087 (1); The numbers move toward the Kaprekar constant 6174 (2).	1+1+1+2

	OR: Tennis: 3 symbols; Cricket: 6 symbols; Kabaddi: 7 symbols (1); Total symbols $= 16$ (1); Total students $= 80$ (1); Write the key: 1 symbol = 5 students (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 120	Triangular number $T_{15} = \frac{15\times16}{2}$.
2	**(b)** 270	Remaining turn $= 360^\circ -$ turn.
3	**(d)** 2	Keep reversing and adding.
4	**(b)** 75	Add all.
5	**(d)** 6	Check each number.
6	**(b)** 256	16×16.
7	**(d)** 6	Each pair of rays makes an angle.
8	**(b)** 22	Add all the digits.
9	**(c)** 2	Count how many times each value appears.
10	**(d)** 5	$72 = 2 \times 2 \times 2 \times 3 \times 3$
11	**(a)** 45	Triangular numbers: term n is $1 + 2 + \cdots + n$.
12	**(d)** 145	Inner and outer scales add up to 180°.
13	**(a)** 31	Add the two cells below.
14	**(b)** 6	Count the values bigger than the given number (not equal to it).
15	**(c)** 3	The next un-crossed number after 2 is 3.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Each point joins to 7 others, and every line is counted twice (1); Lines $= \frac{8\times7}{2} = 28$ (1).	1+1
17	Hour hand points to 3 on the dial; each mark is 30° (1); Angle $= 90^\circ$ (1).	1+1
18	Largest 9621, smallest 1269 (1); Difference $= 8352$ (1).	1+1
19	Height = number $\div 5$ (1); Football: 7 cm; Chess: 3 cm; Hockey: 4 cm (1).	1+1
20	Try prime pairs (1); 5 + 19 = 24 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Follow the rule step by step (1); Sequence: 13, 40, 20, 10, 5, 16, 8, 4, 2, 1 (1); Steps $= 9$, maximum $= 40$ (1).	1+1+1
22	Values: $40, 70, 50$ (1); Largest $= 70$, smallest $= 40$ (1); Difference $= 30$ (1).	1+1+1
23	Split the number into two factors repeatedly (1); $96 = 2 \times 2 \times 2 \times 2 \times 2 \times 3$ (1); Product check: 96 (1).	1+1+1
24	Difference between terms is 2 (1); 10th term $= 5 + 9 \times 2$ (add the difference 9 times) (1); $= 23$ (1).	1+1+1
	OR: One right angle is 90° (1); $180 \div 90 = 2$ (1); 3 right angles make 270° (1).	1+1+1
25	Digit sum $= 7 \times 5 = 35$ (1); Not divisible by 3, because the digit sum is not a multiple of 3 (2).	1+2
	OR: A bar graph should fit on the page, so use the bigger scale (1); Largest value 60 gives 6 cm with 1 cm = 10 (1); Hence choose 1 cm = 10 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Cross out multiples of 2, 3, 5, 7 (1); Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37 (1); Number of primes $= 12$ (1); Check by testing factors (2).	1+1+1+2
	OR: Next term $= 8 + 13 = 21$ (1); Then 34 and 55 (2); Sum of first 5 terms $= 19$ (equals the term two places ahead minus 2: 21-2) (2).	1+2+2
27	The hour hand turns 30° every hour (1); Hours passed $= 4 - 1 = 3$, hour hand turn $= 90^\circ$ (1); Minutes passed $= 180$; the minute hand turns 3 full turns $= 1080^\circ$ (1); Type of the hour hand's turn: right (2).	1+1+1+2

	OR: 121+232=353 (1); Palindrome (1); Start 29: 29, 121 (1); Palindrome 121 after 1 step(s) (2).	1+1+1+2

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 216	Cube numbers are $1^3, 2^3, 3^3, \ldots$
2	**(a)** collinear points	Definition / property
3	**(a)** 19	Middle row: $a+b$ and $b+c$; top $= a+2b+c$.
4	**(d)** 55	Count the half symbol as half the value.
5	**(b)** 3	Test 2, 3, 5, 7, ...
6	**(b)** 34	Each term is the sum of the two terms before it.
7	**(b)** 180	At full hours the minute hand is at 12; each hour mark is 30°.
8	**(c)** 77	Reverse and add.
9	**(d)** frequency	Definition
10	**(a)** 5128	Apply the divisibility rule of 8.
11	**(a)** 6	Number of lines $= \frac{4\times3}{2}$.
12	**(a)** right	Classify the turn.
13	**(b)** 7353	Arrange digits in descending and ascending order and subtract.
14	**(c)** 3.5	Divide by the value of one symbol.
15	**(b)** 14 and 25	Co-prime numbers have no common factor other than 1.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Sum of first n odd numbers is $n \times n$ (1); $13 \times 13 = 169$ (1).	1+1
17	Pairs of points (1); 6 segments (1).	1+1
18	Even: halve it; odd: multiply by 3 and add 1 (1); $6, 3, 10, 5, 16, 8, 4, 2, 1$ (1).	1+1
19	Count each value (1); 1: 3; 2: 5; 3: 2; 4: 4 (1).	1+1
20	Factors of 14: 1, 2, 7, 14; of 15: 1, 3, 5, 15 (1); Co-prime (only common factor is 1) (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	LCM $= 12$ (1); Common multiples: 12, 24, 36, 48 (1); Each is a multiple of the LCM (1).	1+1+1
22	Sum $= 55 + 66 = 121$ (1); $121 = 11 \times 11$ (1); Two consecutive triangular numbers always add to a square number (1).	1+1+1
23	Every 90° is a quarter turn: N to E to S to W (1); 180° gives South (1); Type: straight (1).	1+1+1
24	Repeat the largest-minus-smallest step (1); $4821 \to 7173 \to 6354 \to 3087 \to 8352 \to 6174$ (1); Number of steps $= 5$ (1).	1+1+1
	OR: (i) Orange: 65 (1); (ii) Mango: 45 (1); (iii) total $= 170$ (1).	1+1+1
25	Sum of digits at odd places and even places (1); Difference $= 0$ (1); Divisible by 11 (1).	1+1+1
	OR: $1 + 2 + \cdots + 4 = 10$ (1); $1^3 + \cdots + 4^3 = 1 + 8 + 27 + 64 = 100$ (1); $10^2 = 100$, so they match (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Count each value: 0: 1; 1: 1; 2: 3; 3: 3; 4: 2; 5: 4 (1); Most common: 5 (2); More than 2: 9 students (2).	1+2+2
	OR: 3 and 4 have no common factor (1); Example: $60 \div 3 = 20$ and $\div 4 = 15$ (1); $60 \div 12 = 5$ (1); So both factors divide it and their product also divides (2).	1+1+1+2
27	First person shakes 6 hands, next 5 new hands, ... down to 1 (1); Total $= 6 + 5 + 4 + 3 + 2 + 1$ (1); $= 21$ handshakes (1); This is the triangular number T_6 (2).	1+1+1+2

	OR: $\angle AOC = 135 - 45 = 90^\circ$ (1); $\angle AOB = 45^\circ$: acute (1); $\angle BOC = 135^\circ$: obtuse (1); $\angle AOC = 90^\circ$: right (1); State the arms and vertex (1).	1+1+1+1+1

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